

Roll No.....

END SEMESTER EXAMINATION DEC-2024

Name of the Program: B. Tech Petroleum Engineering

Semester: 7th

Name of the Course: Oil and Gas Field Development Planning and Economics

Course Code: TPE 720

Time: 3 Hrs

Maximum Marks: 100

Notes:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a,b, & c in each main questions
- (iii) Total marks in each main question are twenty.
- (iv) Each questions carries 10 marks.

Q1		(20 marks)	
(a)	Describe the life cycle of an oil and gas field in the context of field development planning.		CO1
(b)	Describe the criterias which are to be considered for field development plan in case of onshore field.		CO1
(c)	Describe by diagram range of ultimate discovery in the life cycle of a field.		CO1
Q2		(20 marks)	
(a)	Describe what factors influence the field development plan?		CO4
(b)	Mention the oil recovery obtained under different types of drive mechanism.		CO2
(c)	What data is collected from initial wells which help in preparing development plan.		CO2
Q3		(20 marks)	
(a)	How does well spacing impact the economic feasibility of a field's development plan?		CO2
(b)	What are the key factors to consider when designing a water injection plan for a field?		CO3
(c)	Define reserves. Explain the classification of oil reserves.		CO2
Q4		(20 marks)	
(a)	Describe the purpose of productivity test in bottom hole studies in oil reservoir.		CO2
(b)	Describe the significane of presure and temperature data in hydrocarbon exploration and production.		CO3
(c)	Explain the estimation of reserve by material balance method.		CO3
Q5		(20 marks)	
(a)	Define sick well? List and desribe the crieria for the identification of sick well.		CO5
(b)	Describe the role of Cement Bond Log in sick well .		CO5
(c)	Mention the oil recovery obtained under different types of drive mechanism.		CO6

